

AYAN CHANDRASEKARAN

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EDUCATION

University of Michigan

B.S. Computer Science (College of Engineering), Minor in Music

Ann Arbor, MI

Expected Graduation May 2028

SAT: 1570 Skills: Python, TypeScript/JavaScript, Java, C/C++, HTML/CSS, SQL, Swift, React, Flutter, PyTorch, LLMs,

Responsible AI, Agentic Systems/MCPs, Git, REST APIs, Agile/Scrum, Digital Signal Processing

Relevant Coursework: Data Structures & Algorithms, OOP [EECS 280/281], Machine Learning [EECS 245], Cybersecurity [EECS 388], Computer Organization/Assembly [EECS 370]

EXPERIENCE

Smith+Nephew

Pittsburgh, PA

Intern, Machine Learning and Computer Vision for Surgical Robotics

Summer 2026 - Presented at company-wide AI Conference

May 2026 – Present

- Built CNN/VLM classification app for image-based Pressure Ulcer staging, diagnosis, and product recommendation
- Trained/fine-tuned a **ResNet-50** classifier with ordinal regression loss, achieving **human clinician-level bedside accuracy (74% vs 67%)** on high-confidence predictions using a single image with **no physical exam or patient history necessary**
- Designed a human-in-the-loop experiment pipeline for the CNN model development– dispatching subagents to implement and test parallel hypotheses against a defined goal– **reducing dangerous under-staging errors by 21%**

Summer 2025

June 2025 – August 2025

- Cleaned/analyzed surgical data from **6,500+ cases across 60 surgeons**, holding interviews and cross-functional meetings
- Trained Gradient Boosted Trees and Probabilistic models to predict the Total-Knee surgery time based on surgeon/patient characteristics

Michigan Music Business Club (MMBC), Director of Creative Technology

Ann Arbor, MI

Director of Creative Technology, using software to create immersive events

January 2025 – Present

- Led production for real-time visuals for annual **conference (500+ attendees**, paid event featuring music label CEOs and industry executives), running large-screen background visuals during speaker sessions
- Organized **concerts** and artist **showcases**– coordinating visuals, audio setup, stage rigging, and managing budget
- Built Artist LaunchPad **website** and assisted in Conference iOS/Android **app** to help attendees plan which events to visit

Google Student Developers Club @ Michigan, Project Manager

Ann Arbor, MI

Project Manager, building software solutions for charity organizations

October 2025 – Present

- Led wellness-check system for **HomeRise**, a shelter network with **18 locations across San Francisco**, by building an **automated alert pipeline** in Python that notifies staff when a resident hasn't checked in within 72 hours
- Designed the check-in form and submission pipeline (HTML/CSS/JS, form validation, backend intake) feeding the alert system, **coordinating with an 8-person team** to ship the feature into production
- Built an end-to-end audio recording and transcription pipeline in Flutter for a mental-health-focused audio journaling app

Emanate, Startup (<https://www.emanate.studio/>)

Ann Arbor, MI

Founder, producing interactive visuals for concerts and events on campus

August 2025 – Present

- Developed C++/Python software enabling musicians to generate audio with **granular synthesis** using [hand gestures](#)
- Implemented a [projection mapping](#) system for musical installations on campus (official installation @ **2026 MMBC Conference**), including a **playable projected piano**

PROJECTS

synqd, iOS App (<https://mysynqd.com/>) [Full-Stack, Swift, SQL, Version Control]

Remote

Founder, notification-based long-distance couples app to see current timezone and send custom messages

May 2026 – Present

- Built iOS app on **Supabase** backend, keeping home/lock screen widgets in sync with a partner's **live timezone/location**
- Engineered **real-time sync backend** (Postgres, edge functions, APN) that pushes partner location updates in **under 1 second**
- Drove development with **parallel AI agents using Git worktrees**, directing them to write tests and docs using custom skills
- Launched on TestFlight to **30 users in the first week** and shipped their requests across 4+ builds

SOLO.AI, iOS App [Swift, Python, On-Device ML, Local AI/Agent]

Remote

Founder, privacy-first iOS chat app that scrubs personal data on-device before any cloud LLM sees it

April 2026 – Present

- Architected **hybrid inference pipeline** running a local language model fully on-device via a custom Swift package, **cutting response start time from ~20s to ~2.7s**
- Designed a **three-tier privacy system** where the on-device model detects Personally Identifiable Information (PII), **reformulates sensitive prompts** to minimum necessary context, and **escalates high-risk** ones to user review
- Implemented **streaming responses** and **local voice input**, validating the privacy system with an **89-test automated suite**

ACTIVITIES & LEADERSHIP

Y Combinator Startup School

San Francisco, CA

Accepted from 30,000+ applicants

August 2026

Journal of Emerging Investigators

Chicago, IL

[Published Research](#)

June 2021 – August 2021